

Final Project – Proposal

Project Title: Out For Music

Overview

Out For Music is a web application that helps users discover new music by searching for artists, albums, and songs. Using the Spotify Web API, users can browse artist profiles, album information, track listings, release dates, cover art, and listen to available song previews. The application also integrates with the Mercado Libre API, allowing users to search for physical copies of an album, such as CDs and vinyl records.

The problem this application solves is that music fans often use multiple websites to discover music and then search for places to purchase physical albums. This application combines both experiences into a single, easy-to-use platform. My motivation for this project comes from my own interest in music and collecting physical albums, making it a project that is both useful and enjoyable to build.

Target Audience

This application is designed for: Music enthusiasts, vinyl and CD collectors, casual listeners looking for new artists, students and young adults and anyone interested in discovering and purchasing music

Major Functions

1. Search Artists

Users can search for artists by name using the Spotify API.

2. View Artist Information

Display artist images, genres, popularity score, and followers.

3. Browse Albums

View all available albums for a selected artist.

4. Album Details

Display:

- Album cover
- Release date
- Total tracks
- Album type
- Track list

5. Song Preview

Allow users to listen to Spotify preview clips when available.

6. Search Album on Mercado Libre

Generate a search using the album and artist name and display available listings.

7. Favorites

Allow users to save favorite albums in Local Storage.

8. Responsive Design

Provide an optimized experience on desktop, tablet, and mobile devices.

Wireframes

Major pages:

Home

- Search bar
- Popular artists
- Featured albums
- Navigation

Artist Page

- Artist image
- Genres
- Popularity
- Albums

Album Details

- Cover image
- Track list
- Release date
- Preview player
- Button:
 - Search on Mercado Libre

Favorites

- Saved albums
- Remove favorite button

External Data

Spotify Web API

Used for:

- Artists
- Albums
- Tracks
- Album covers
- Genres
- Popularity
- Audio previews

Mercado Libre API

Used for:

- Album listings
- Prices
- Images
- Seller information
- Purchase links

Data Stored

Using Local Storage:

- Favorite albums
- Recently searched artists

- User preferences (optional)
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Module List

api.js

Handles communication with Spotify and Mercado Libre APIs.

search.js

Processes search requests.

artist.js

Displays artist information.

album.js

Displays album details.

mercadoLibre.js

Searches Mercado Libre for albums.

favorites.js

Stores and manages favorite albums.

storage.js

Handles Local Storage.

ui.js

Updates the user interface.

main.js

Initializes the application.

Graphic Identity

Color Palette

Primary

- #7B61FF (Royal Purple)

Secondary

- #1E1B3A (Midnight Indigo)

Accent

- #FFFFFF (Pure White)

Background

- #0F1226 (Deep Navy)

Highlight

- #FF9F6E (Soft Coral)

Optional Action Color

- #3CCFCF (Bright Turquoise)
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Typography

Headings

- Poppins

Body

- Roboto
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Style

- Rounded buttons
 - Card-based layout
 - Large album artwork
 - Dark theme inspired by music streaming platforms
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Application Icon

A vinyl record with the letters OFM at the center.

Timeline

Week 5

- Create project structure
- Build navigation
- Connect Spotify API
- Implement search functionality
- Display artists

Week 6

- Display albums
- Album detail page
- Favorites using Local Storage
- Responsive layout
- Improve UI

Week 7

- Integrate Mercado Libre API
 - Error handling
 - Testing
 - Final polishing
 - Deploy project
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Project Planning (Trello)

<https://trello.com/b/NkXaLiJ/final-project>

Challenges

The biggest challenge will likely be integrating two different APIs with different authentication methods and response formats. Spotify requires an access token and OAuth authentication, while Mercado Libre returns marketplace search results with a different data structure. Another challenge will be handling cases where song previews or album listings are unavailable. Ensuring a responsive, user-friendly interface across different screen sizes and implementing efficient error handling for failed API requests will also require careful planning. Finally, managing asynchronous API calls and presenting the data in a clear and organized way will be an important aspect of the application's development.